Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	15
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2576	3013	1380.42	px^2
major_axis_length_px	lacuna shape	109.567	124.464	38.2072	px
minor_axis_length_px	lacuna shape	31.8891	33.1799	10.1914	px
aspect_ratio	lacuna shape	3.6142	3.6285	1.2579	unitless
eccentricity	lacuna shape	0.9409	0.9613	0.0498	unitless
solidity	lacuna shape	0.7493	0.7672	0.1286	unitless
roots_count	headline	15.1333	16	6.8958	count
ring_length_r30_px	headline	485.533	459	197.359	px
ring_length_r60_px	reported	1045.73	992	448.837	px
owned_length_px	ownership-dependent	2175.82	2113.87	1373.09	px
edge_count	ownership-dependent	99.9333	92	64.7373	count
mean_edge_length_px	ownership-dependent	21.7572	21.6373	3.0691	px
perimeter_px	normalised	292.07	309.522	122.808	px
ring_area_r30_px2	normalised	10491.5	11337	3304.03	px^2
ring_area_r60_px2	normalised	24400.9	26961	7815.16	px^2
in_frame_fraction_r30	normalised	1	1	0.0001	unitless
in_frame_fraction_r60	normalised	0.9858	1	0.0377	unitless
ring_density_r30	normalised	0.045686	0.0456894	0.00665998	px^-1
ring_density_r60	normalised	0.0418245	0.0403269	0.00704682	px^-1
roots_per_100px_perimeter	normalised	5.546	5.2653	1.7789	per 100 px
ring_attached_length_r30_px	network v2	445.267	404	197.604	px
ring_attached_length_r60_px	network v2	892.733	690	461.08	px
ring_length_w_r30_px	network v2	552.583	514.718	228.616	px
ring_length_w_r60_px	network v2	1187.57	1109.92	515.971	px
sholl_crossings_r10	network v2	10.7333	11	5.2026	count
sholl_crossings_r20	network v2	13.2667	13	5.3381	count
sholl_crossings_r30	network v2	13.6667	12	5.5119	count
loop_count	loops, report only	8.2	5	8.0196	count
loop_edge_share	loops, report only	0.2613	0.2283	0.1204	share
ring_width_mean_r30_px	width	7.6454	7.62	0.4946	px

Per field
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measure	value	units
analysed_area_px2	1.0064e+06	px^2
total_skeleton_length_px	40198	px
canalicular_length_density_per_px	0.0399422	px^-1
junction_count	934	count
junction_density_per_px2	0.00092806	px^-2
lacuna_count	19	count

lacunae_per_px2	1.888e-05	px^-2
skeleton_component_count	227	count
mean_component_length_px	177.084	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0452403	px^-1
field_density_without_flagged_per_px	0.0398259	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	355	px
unattached_component_count	19	count
canalicular_length_density_attached_per_px	0.0399422	px^-1
field_length_density_with_unattached_per_px	0.040295	px^-1
loop_count	159	count
loop_edge_share	0.3259	share
edge_count	2007	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2188	px
width_median_px	8.2462	px
width_p10_px	5.6569	px
width_p90_px	10.7703	px
width_pixels_used	38859	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	445	56.9837	11.5357	4.9398	0.9793	0.7672	-0.2356	148.654	554.791	False
3	1001	62.1533	28.1242	2.21	0.8918	0.5533	-0.1968	55.7832	107.949	False
4	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
5	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
6	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
7	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
8	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
9	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
10	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
11	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
12	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
13	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
14	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
15	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
16	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True
17								310.754	226.539	False
18								933.644	3.9103	True
19								368.378	576.748	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	7	218	587	843.269	34	24.802	7.8744
2	8	278	584	906.583	40	22.6646	8.655
3	14	376	696	1457.74	53	27.5045	7.6517
4	12	317	716	1201.8	52	23.1116	7.0968
5	10	462	1118	3418.7	158	21.6373	7.7589
6	29	830	1707	4818.15	205	23.5032	7.6195
7	16	421	760	890.068	40	22.2517	7.3807
8	17	459	882	919.796	44	20.9045	7.5303
9	6	412	992	2113.87	99	21.3522	8.6237
10	16	514	1313	3980.45	160	24.8778	7.62
11	24	870	1902	4169.76	231	18.0509	7.3977
12	23	613	1351	2381.13	114	20.8871	7.0375
13	20	608	1318	2292.4	121	18.9454	7.7263
14	10	317	719	2320.32	92	25.2209	6.9414
15	8	378	910	2468.41	95	25.9833	7.6393
16	5	192	542	200.167	9	22.2407	6.5763
17	17	638	1378	1557.2	76	20.4894	7.9873
18	2	87	223	413.191	20	20.6595	8.3221
19	5	168	250	209.397	14	14.9569	7.6535

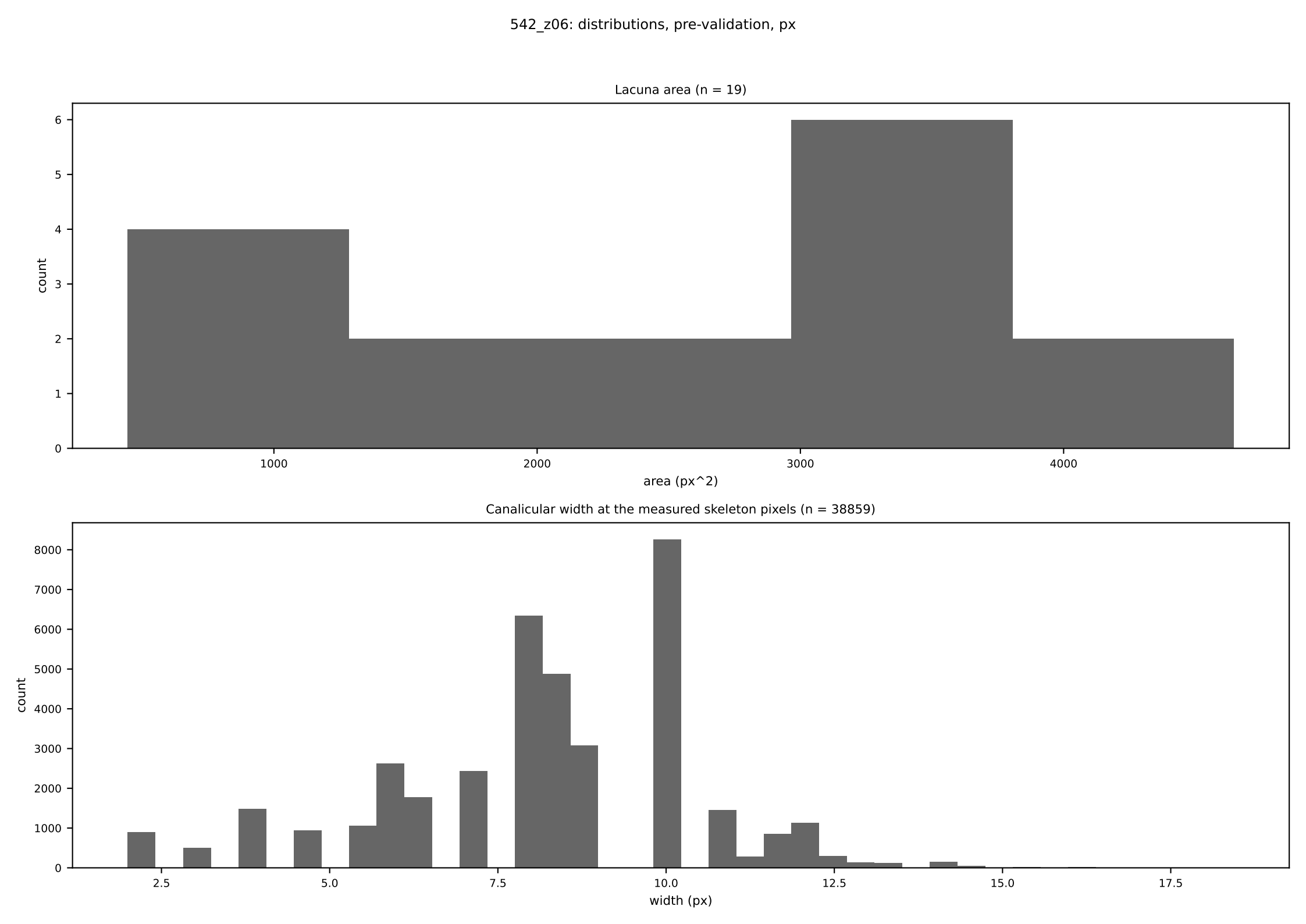
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0388107	0.0462423	3.2523
2	126.468	6226	14794	1	1	0.0446515	0.0394755	6.3257
3	225.309	8379	19327	0.9998	0.884	0.0448741	0.0360118	6.2137
4	151.782	6752	15321	1	1	0.046949	0.0467332	7.9061
5	322.043	11663	28685	1	1	0.0396125	0.0389751	3.1052
6	438.517	13901	31616	1	1	0.0597079	0.0539917	6.6132
7	211.924	8423	18846	1	1	0.0499822	0.0403269	7.5499
8	359.191	12383	26961	1	0.9025	0.037067	0.0327139	4.7329
9	302.451	10940	24775	1	1	0.03766	0.0400404	1.9838
10	309.522	11337	28286	1	1	0.0453383	0.0464187	5.1693
11	468.274	15363	34096	1	1	0.0566296	0.0557837	5.1252
12	422.978	13165	30001	1	1	0.0465629	0.0450318	5.4376
13	379.848	13296	32108	1	1	0.045728	0.041049	5.2653
14	217.924	8830	22986	1	1	0.0359003	0.0312799	4.5888
15	332.894	10505	23806	0.8602	0.7919	0.0359829	0.0382257	2.4032
16	205.776	6766	16357	0.7841	0.714	0.0283772	0.0331357	2.4298
17	387.948	13037	31120	1	1	0.0489376	0.0442802	4.382
18	87.0061	2915	8653	0.5433	0.5282	0.0298456	0.0257714	2.2987
19	56.8701	3677	7091	1	1	0.0456894	0.035256	8.792

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	218	580	251.48	670.115	6	5	5
2	259	453	316.179	657.973	5	8	8
3	316	641	432.404	791.654	8	7	10
4	313	607	356.179	803.056	6	7	9
5	404	958	537.316	1286.86	9	16	15
6	827	1612	956.264	1962.63	22	20	22
7	360	556	474.433	859.683	8	13	11
8	415	690	514.718	986.838	11	11	10
9	357	624	462.747	1109.92	12	13	12
10	500	1290	590.316	1506.33	11	14	16
11	801	1825	997.566	2153.17	17	22	25

12	561	1169	692.257	1529.35	16	15	15
13	575	1220	688.671	1491.87	15	22	19
14	302	622	355.522	810.713	7	9	9
15	351	746	427.676	1026.47	12	11	16
16	176	205	217.238	605.073	6	7	7
17	583	1002	730.47	1588.01	12	16	18
18	62	169	97.7696	249.338	2	3	5
19	106	122	183.711	275.48	2	6	6


542_z06: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
1
2
3
4
5
6
1000
2000
3000
4000
Canalicular width at the measured skeleton pixels (n = 38859)
count
width (px)
0
1000
2000
3000
4000
5000
6000
7000
8000
2.5
5.0
7.5
10.0
12.5
15.0
17.5